ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25
DIPLOMA (CSE/ME-A/ME-P/CE/EE)
Semester: II
SUBJECT CODE: DME201
FUNDAMENTALS OF MECHANICS

Time: 03:00 Hrs. **Maximum Marks: 60**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. **(12 Marks)**

Q. 1. Answer the following questions: **(1×6 = 6 Marks)**

(i) Define coplanar and non-coplanar force systems.
(ii) What is the moment of inertia? State its unit.
(iii) State Lami's theorem
(iv) Differentiate between static and kinetic friction.
(v) Define moment of a force.
(vi) Angle of Friction

Q. 2. Answer the following questions: **(1×6 = 6 Marks)**

(i) Define Co-efficient of Friction.
(ii) Differentiate between couple and force.
(iii) Define Limiting Force of Friction.
(iv) Define the equilibrium of forces.
(v) Define first law of newton.
(vi) What is the triangle law of forces?

SECTION – B

NOTE:Attempt all questions. **(6×2×2 = 24 Marks)**

Q.3.(i)(a) What is the moment of inertia? Derive the expression for the moment of inertia of a rectangular section about its centroidal axis.

OR

(b) A force of 500 N is acting at an angle of 30° with the horizontal. Find its horizontal and vertical components.

(ii)(a) Explain the different types of forces with examples.

OR

(b) A body weighing 200 N rests on an inclined plane making an angle of 30° with the horizontal. Find the force required to move the body up the plane if the coefficient of friction is 0.3.

Q.4.(i)(a) Explain D'Alembert's principle and apply it to a simple problem.

OR

(b) The following forces act a point (i) 200 N inclined at 30° towards the North of East. (ii) 250 N towards North. Find the resultant of the force system.

(ii)(a) State the laws of friction and also explain angle of friction.

OR

(b) Resolve the 100N force acting 30° to horizontal into two components, one along horizontal and other along 120° to horizontal.

SECTION – C

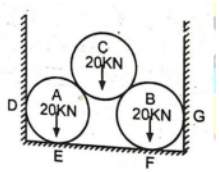
NOTE: Attempt all questions. **(12×2 = 24 Marks)**

Q.5.(a) Prove the parallel axis theorem in the determination of moment of inertia of areas with the help of a neat sketch.

OR

(b) What are various types of loadings? Distinguish clearly between uniformly distributed loads, uniformly varying load.

Q.6.(a) Three smooth pipes each weighing 20 kN and of diameter 60 cm are to be placed in a rectangular channel with horizontal base as shown in the figure. Calculate the reactions at the point of contact between the pipes and between the channel and the pipes. Take the width of the channel as 160 cm.



OR

(b) An electric light fixture weighing 150 N hangs from a point C, by two strings AC and BC as shown in the figure. Determine the forces in the strings AC and BC.

